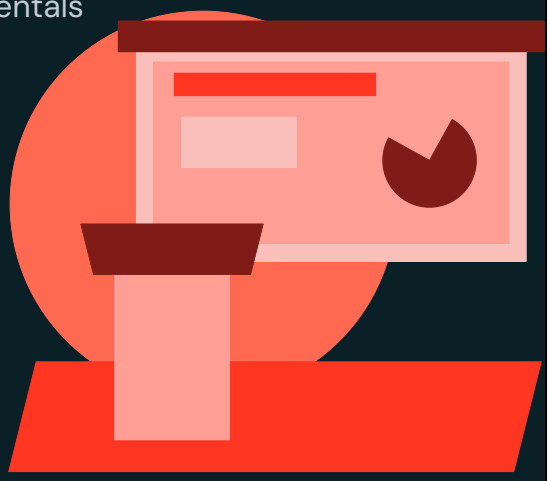
databricks

Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

Simplified Pipeline Development



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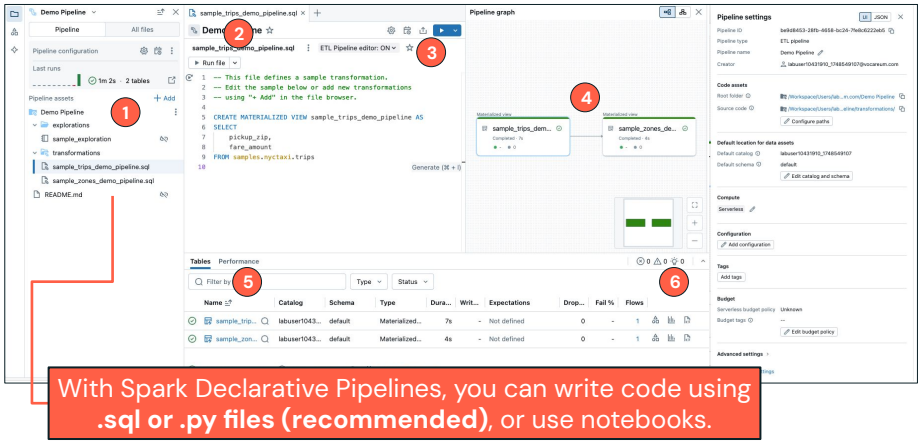
In this lecture, you will learn how the multi-file editor in Lakeflow Declarative Pipelines streamlines ETL pipeline development and debugging with features such as a pipeline asset browser, multi-file code editing, interactive DAG visualization, data previews, execution insights, easier debugging tools, and fast pipeline validation.

Simplified Pipeline Development

Multi-file editor in Lakeflow Spark Declarative Pipelines



With Spark Declarative Pipelines, you can write code using **.sql or .py files (recommended)**, or use notebooks.

What's New

1. Pipeline asset browser
2. Multi-file code editor with features for step-by-step development
3. Pipeline-specific toolbars
4. Interactive DAG
5. Data previews
6. Execution insights panels
   - Easier debugging
   - Faster validation with dry run

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The multi-file editor in Lakeflow Spark Declarative Pipelines makes developing and debugging your ETL pipelines easier and more efficient.

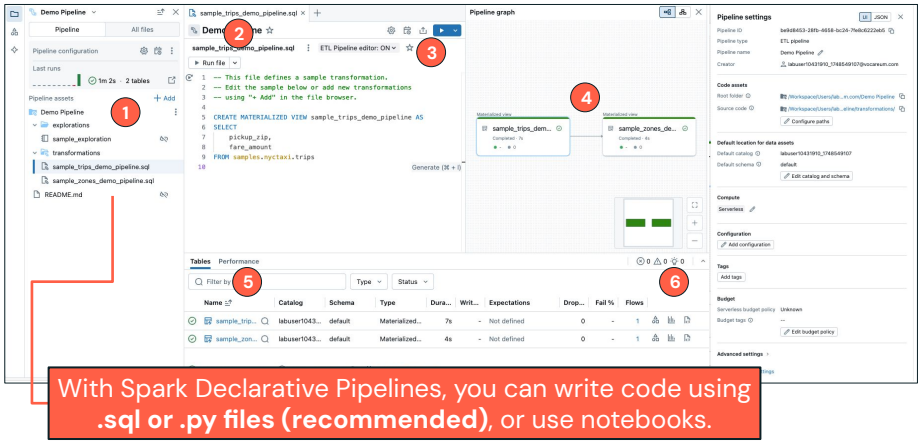
Instead of managing one large file, your pipeline is organized as a set of files visible in the pipeline assets browser, letting you edit code and configure your pipeline components all in one place.

Key new features include:

- Pipeline asset browser to easily navigate pipeline files
- A multi-file code editor designed for step-by-step pipeline development
- Pipeline-specific toolbars for quick access to common actions
- An interactive DAG (Directed Acyclic Graph) for visualizing pipeline dependencies

Simplified Pipeline Development

Multi-file editor in Lakeflow Spark Declarative Pipelines



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- Data previews to inspect intermediate results
- Execution insights panels to monitor pipeline runs
- Easier debugging with integrated tools
- Faster validation through dry run capabilities, so you can check your pipeline without running full executions
- These features streamline your workflow and help you build robust pipelines faster.

Refer to: Develop and debug ETL pipelines with the Lakeflow Pipelines Editor | Databricks on AWS

Additional Features Overview

What's Next: Features Outside This Course

Full UC Support

- Publish to multiple catalogs & schemas
- Read **STs** and **MVs** in Dedicated Access Mode
- Create a Unity Catalog pipeline by cloning a Hive metastore pipeline

Better Performance

Improved performance throughout Lakeflow Spark Declarative Pipelines!

- Serverless
- Incremental refresh for materialized views
- Photon

Databricks Asset Bundles (DABs)

DABs enable you to **programmatically** validate, deploy, and run Databricks resources such as pipelines for **CI/CD production workloads**

- Automated Deployment with Databricks Asset Bundles course

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Deployment with Databricks Asset Bundles (DABs)

DABs enable you to **programmatically** validate, deploy, and run Databricks resources such as pipelines for **CI/CD production workloads**
Refer to:

- What are Databricks Asset Bundles? | Databricks on AWS
- Automated Deployment with Databricks Asset Bundles | Databricks